参数投影器与 Hybrid 三级溯源验证报告

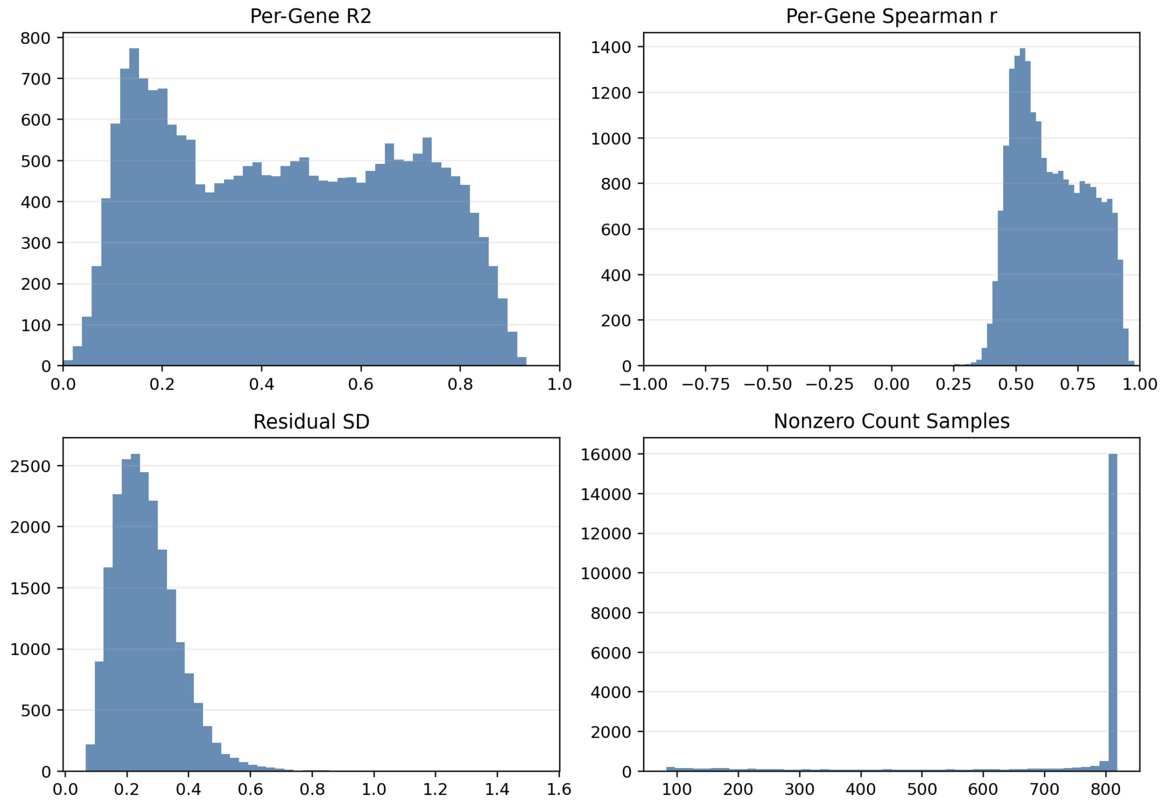
Reference-projected VSD + logCPM local rerank validation summary

项目	内容
主线路线	projected VSD Network Top3 beam + logCPM resolution/local exact rerank
内部验证	Bo2023 LOSO / LOMO / exact region / local rerank / formal three-tier
外部验证	AHBA human RNA-seq; TCGA/BraTS MRI-labeled glioma
输出目录	D:\Download\cfrna-brain-tracing-streamlit-cloud-ready\output\pdf

核心结论：projected VSD 单独用于 exact-region 会损失精度；但作为 Network Top3 beam 与 logCPM 的 resolution/local exact rerank 组合后，内部 LOSO exact Top3 达到 0.4533，外部 AHBA exact Top3 达到 0.4286，是当前最稳健的跨域主线候选。

1. 数据审计与投影器质量

指标	结果
Bo2023 common samples	819
Common gene symbols	NA
Locked network genes present	NA/200
Projector median sample Pearson	NA
Projector global Pearson	NA
Projector MAE	NA
Median per-gene R2	NA
Date-like gene symbols after cleaning	0

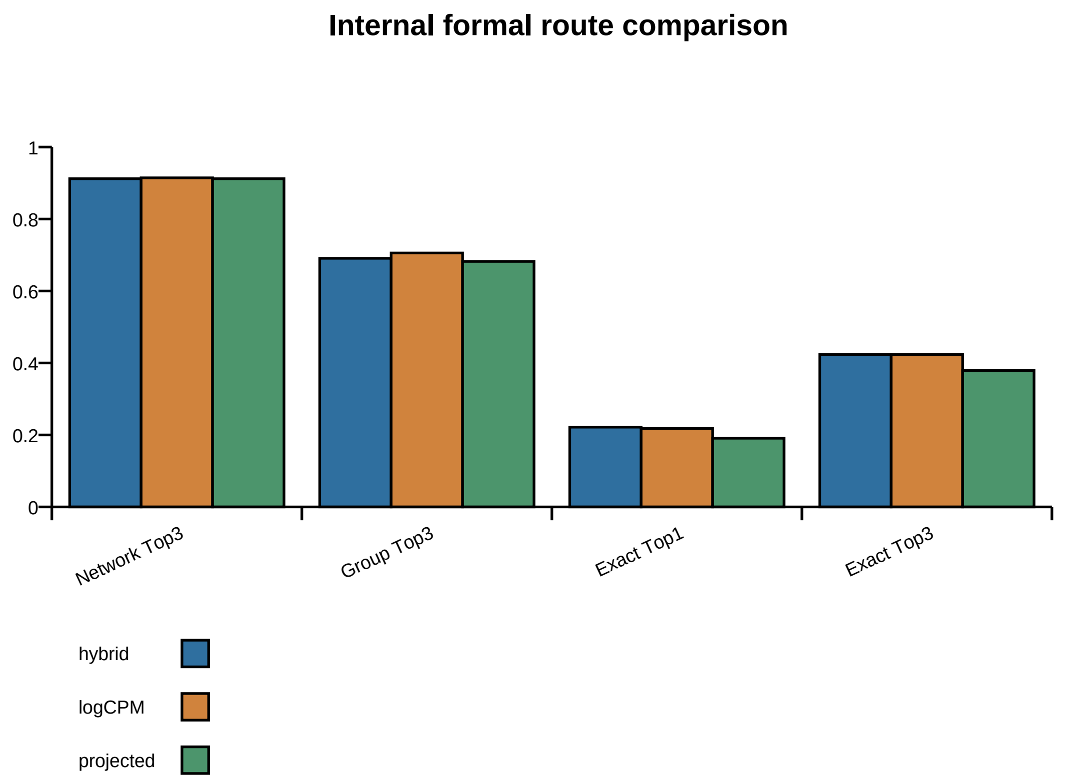


解释：Excel 日期样式基因符号已清理；剩余 ENSMFAG fallback 主要是无稳定 human-like symbol 的基因，不影响 locked 200 network gene 覆盖，但外部人类数据 exact 层应避免依赖这些 fallback gene。

2. 内部验证总览

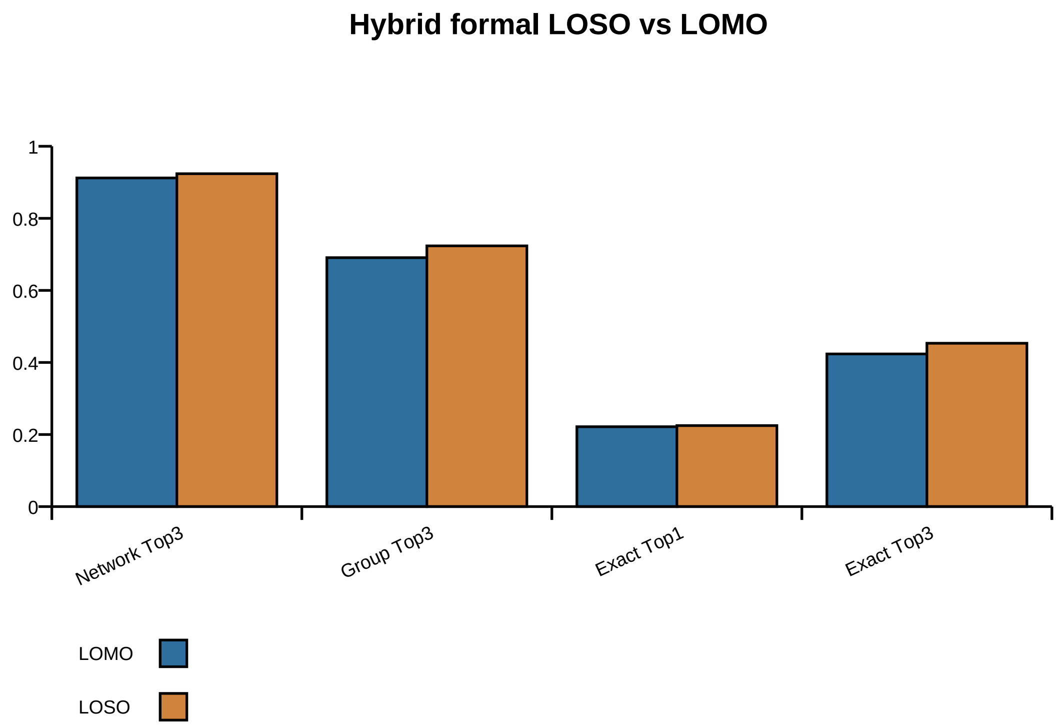
本节按从简单到正式的顺序汇总：Network LOSO/LOMO、direct exact、local rerank、完整三级 hybrid。

验证	route	Top1	Top3
LOSO network	logcpm_baseline	0.5556	0.8828
LOSO network	native_vsd	0.5311	0.8816
LOSO network	projected_vsd	0.5800	0.9158
LOMO network	logcpm_baseline	0.4628	0.8510
LOMO network	native_vsd	0.5043	0.8718
LOMO network	projected_vsd	0.5372	0.9133



3. 完整三级 Hybrid: LOMO 与 LOSO

验证	Network Top1	Network Top3	Group Top1	Group Top3	Exact Top1	Exact Top3
LOMO hybrid	0.5775	0.9121	0.4138	0.6909	0.2217	0.4236
LOSO hybrid	0.5835	0.9238	0.4447	0.7236	0.2248	0.4533



完整三级 LOSO 是本线程最新补齐的验证：814 个可评估 fold，Exact Top3=0.4533，Group Top3=0.7236，强于 LOMO 数值；这是当前内部最支持 hybrid 主线的证据。

4. Exact-region 与 local rerank 证据链

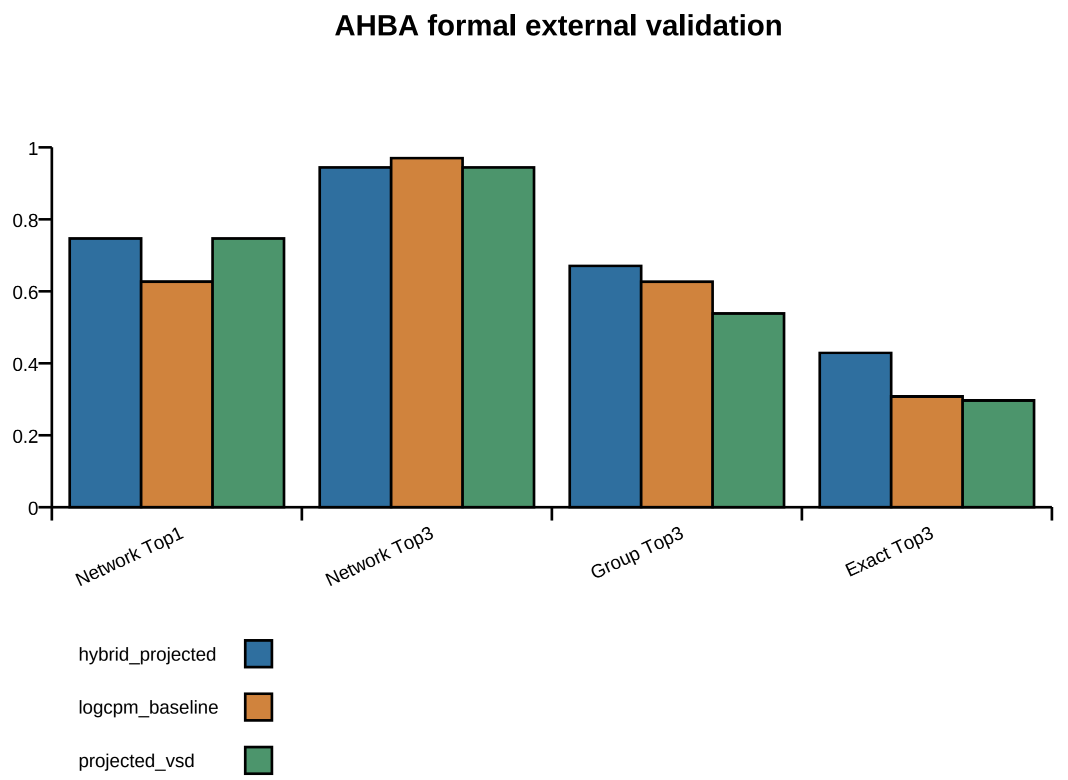
验证	route	Exact Top1	Exact Top3	备注
Direct exact LOSO	logcpm_baseline	0.1376	0.2604	locked 200 direct
Direct exact LOSO	native_vsd	0.1671	0.3403	locked 200 direct
Direct exact LOSO	projected_vsd	0.1671	0.3563	locked 200 direct
Local rerank LOSO	hybrid projected network logcpm local region genes	0.2088	0.4263	Top3 network beam + local genes
Local rerank LOSO	logcpm top3 network local region genes	0.2101	0.4263	Top3 network beam + local genes
Local rerank LOSO	native vsd top3 network local region genes	0.2088	0.4238	Top3 network beam + local genes
Local rerank LOSO	projected vsd top3 network local region genes	0.2162	0.4337	Top3 network beam + local genes

趋势：direct exact 最弱；加入 network beam 和 region-local rerank 后 exact Top3 明显提升；完整三级再加入 resolution group 和 Top50/Top100 fusion 后，hybrid LOSO exact Top3 进一步升到 0.4533。

5. 外部验证: AHBA 与 TCGA/BraTS

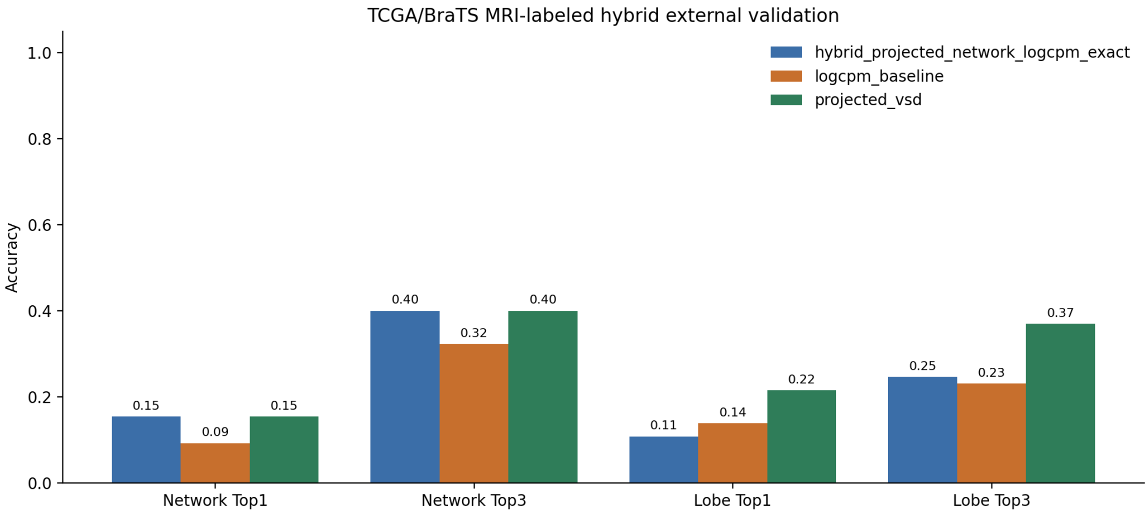
AHBA human RNA-seq formal three-tier

route	Network Top1	Network Top3	Group Top3	Exact Top1	Exact Top3
hybrid_projected_network_logcpm_exact	0.7468	0.9442	0.6703	0.2418	0.4286
logcpm_baseline	0.6266	0.9700	0.6264	0.1758	0.3077
projected_vsd	0.7468	0.9442	0.5385	0.1099	0.2967



TCGA/BraTS MRI-labeled glioma

route	Network Top1	Network Top3	Lobe Top1	Lobe Top3	Broad Top3
hybrid_projected_network_logcpm_exact	0.1538	0.4000	0.1077	0.2462	0.6462
logcpm_baseline	0.0923	0.3231	0.1385	0.2308	0.2462
projected_vsd	0.1538	0.4000	0.2154	0.3692	0.6462



注意：TCGA/BraTS 的 MRI truth 是 human atlas labels，不是 Bo2023 macaque region ID，因此报告 network/lobe/broad accuracy，不报告 Bo2023 exact-region accuracy。

6. 结论与下一步

结论	证据
projected VSD 适合作为 Network Top3 beam	内部 LOSO Top3=0.9238；AHBA/TCGA 外部保留 network/broad 优势
纯 projected VSD 不适合作为 exact 主路线	formal LOMO pure projected Exact Top3=0.3793，低于 hybrid/logCPM
hybrid 是当前跨域主线候选	内部完整三级 LOSO Exact Top3=0.4533；AHBA Exact Top3=0.4286
logCPM 仍是强内部基线	formal LOMO logCPM Group Top3=0.7057，略高于 hybrid=0.6909
外部 exact 需按标签类型解释	AHBA 可做 mapped exact；TCGA/BraTS 只能做 network/lobe/broad

推荐主线：projected VSD Network Top3 beam + logCPM resolution/local exact rerank。短期下一步应固定该路线为外部验证默认路线，并对 AHBA/TCGA 的误差标签做按类别审计；长期应加入人-猕猴 region label 映射层，减少 exact-region 的跨物种标签偏差。